-While NNLM, Word2Vec and GloVe generate good embedding vectors, it's difficult to apply them to rich language. To overcome this FastTex was created.

-It learns representations for character n-grams

-The key idea is to represent a word as a subword to create the word vector. Meaning in represents a word by the sum of the vector representations of its n-grams.

-Word embedding vector spaces demonstrates how close words are to each other and observe how semantically similar words cluster together.

-We can form word embedding spaces with 2 different languages, having the same words close to each other. We can also make word embedding with images, having photo's close to what they represent in the English language.

-t-SNE allows visualization of document embedding using full sentences instead of simple words. Putting sentences closer to each other based on their toppings.

-Distributed memory (PV-DM) models use the beginning of the sentence(word vector) and the paragraph id to predict the rest of the sentence.

-The second method is the distributed bag of words (PV-DBOW), it uses only the paragraph id to predict the sentences, it ignores the context words in the input and forces the model to predict words randomly sampled from the paragraph in the output.

-Combination of PV-DM and PV-DBOW are recommended

-Supervised paragraph vector (SPV) is used for class embedding, but can also be used to calculate reviews, scaling words between -1 and 1 depending on how negative and positive they are.